



ANDERSON JUNIOR COLLEGE
2015 JC 2 PRELIMINARY EXAMINATIONS

NAME: _____

PDG: _____ /14

CHEMISTRY

Higher 2

Paper 2

9647/02

1 September 2015

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use				
Paper 2	1	/ 12	Total	/ 72
	2	/ 14		
	3	/ 12		
	4	/ 12		
	5	/ 9		
	6	/ 13		

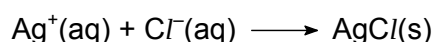
This document consists of **19** printed pages and **1** blank page.

1 Planning (P)

FA 1 is a Group II metal chloride hydrate, $MCl_2 \cdot nH_2O$. In order to determine identity of M and the value of n , two different methods – volumetric and gravimetric analysis were carried out. The volumetric analysis depends on an application of the principles of solubility product, while gravimetric analysis is used to measure the mass loss on forming the anhydrous metal chloride.

(a) Volumetric analysis

By titrating a solution of **FA 1** with aqueous silver nitrate, the molar mass of the metal chloride hydrate, $MCl_2 \cdot nH_2O$ can be determined. The equation for the reaction is shown below.



Data for use in this question are given in Table 1.

	colour	K_{sp} (at 25°C)
AgCl	white	$2.02 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$
Ag ₂ CrO ₄	red	$3.01 \times 10^{-12} \text{ mol}^3 \text{ dm}^{-12}$

Table 1

The following reagents are provided.

FA 1 is the Group II metal chloride hydrate, $MCl_2 \cdot nH_2O(s)$

FA 2 is $0.200 \text{ mol dm}^{-3}$ silver nitrate

FA 3 is neutral chromate indicator, $K_2CrO_4(aq)$

The following procedure was carried out.

- Step 1** Weigh 6.002 g of **FA 1** using a weighing bottle and transfer the contents completely into a 100 cm^3 beaker.
- Step 2** Add deionised water to the beaker and stir until the solid has dissolved.
- Step 3** Transfer the contents and washings carefully into a 250 cm^3 volumetric flask and top up the volumetric flask with deionised water to the 250 cm^3 mark. This solution is labelled as **FA 4**.
- Step 4** Use a pipette to transfer 25.0 cm^3 of **FA 4** into a conical flask.
- Step 5** Titrate the solution with **FA 2** using 10 drops of **FA 3** indicator.

- (i) Aqueous silver nitrate is separately added to solutions containing $0.1 \text{ mol dm}^{-3} \text{ Cl}^{-}(\text{aq})$ and $0.01 \text{ mol dm}^{-3} \text{ CrO}_4^{2-}(\text{aq})$. Calculate the concentration of Ag^{+} must be present to cause the precipitation of

AgCl ,

.....

Ag_2CrO_4 .

.....
[2]

- (ii) Using the data given in Table 1 and your answer to (a)(i), suggest what you would see at the beginning of the titration and at the end-point and explain why $\text{K}_2\text{CrO}_4(\text{aq})$ can be used as an indicator in this titration.

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[2]

- (iii) **FA 3** contains a neutral solution of $\text{K}_2\text{CrO}_4(\text{aq})$. Suggest a reason why the titration cannot be carried out in an alkaline condition.

.....
[1]

- (iv) In the experiment, 24.60 cm^3 of **FA 2** was used to completely react with 25.0 cm^3 of **FA 4**.

Outline how you would use this result to determine the molar mass of $\text{MCl}_2 \cdot n\text{H}_2\text{O}$.

.....

[3]

5
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- 2 Magnesium is an abundant mineral in the body. It is naturally present in many food substances. A study by Harvard University discovered that patients who consumed 220 mg of magnesium were 33% more likely to develop diabetes than those consuming 340 mg of magnesium a day.

The Health Promotion Board recommends a daily intake of 300 mg of magnesium a day in order to stay healthy. The magnesium content of some fruits and vegetables that are commonly found in the supermarket are given in the table below.

Fruit / Vegetable	Magnesium Content / mg
1 banana	32
1 cup of low-fat yogurt	43
1 slice of wheat bread	40
1 baked potato with skin	55
1 cup of brown rice	84

- (a) John took 4 slices of wheat bread for breakfast.

Calculate the minimum number of potatoes that he must consume for the rest of the day in order to meet the recommended daily intake of magnesium.

.....

[1]

- (b) A student measured 0.5 cup of brown rice and crushed it in a pestle and mortar to form a powder. The magnesium in the brown rice was then reacted with 0.016 g of nitrogen gas at a very high temperature, producing a yellow solid, **A**.

- (i) Calculate the empirical formula of **A**.

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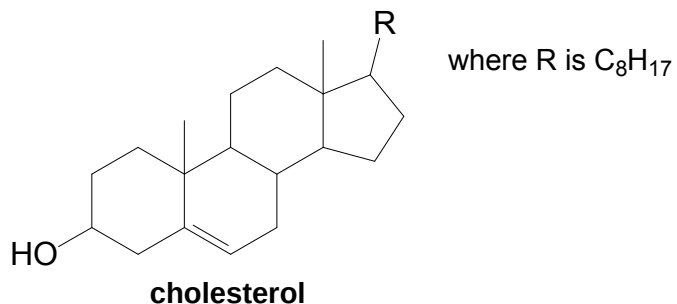
[1]

- (ii) Suggest why a high temperature is required for this reaction.

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[1]

Sufficient level of magnesium is also needed to prevent heart attacks, which occur due to the build-up of plaque that clogs coronary arteries. One of the constituent of plaque is cholesterol.

The structure of cholesterol molecule is



(c) Draw the structural formula of the product formed when cholesterol reacts with

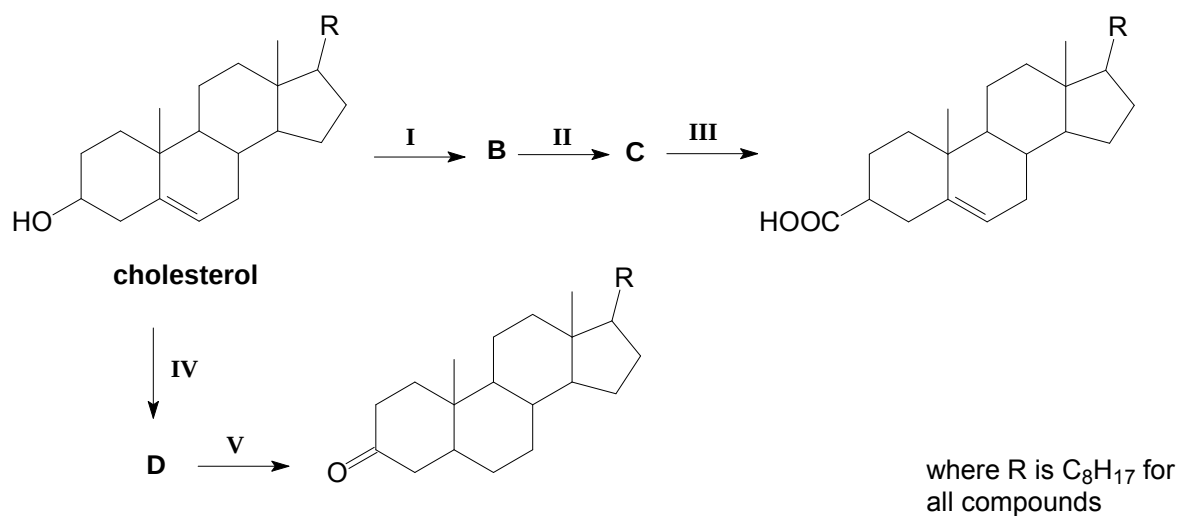
(i) aqueous bromine

[1]

(ii) sodium bromide and concentrated sulfuric acid, high temperature

[1]

(d) Two reaction sequences of cholesterol are shown below.



(i) Suggest a structure for each of the compounds **B**, **C** and **D** in the reaction sequences.

B	C	D
----------	----------	----------

[3]

(ii) State the reagents and conditions required for stages II, IV and V.

Stage II

Reagent:

Condition:

Stage IV

Reagent:

Condition:

Stage V

Reagent:

Condition:

[3]

- (e) Calcium is another Group II metal which is essential for health.

Describe the reactions, if any, that occur when separate samples of calcium oxide and calcium chloride are added to water. Include the approximate pH value of any resulting solutions, and write equations where appropriate.

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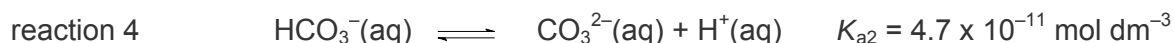
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.....[3]

[Total: 14]

- 3 Decaying of organic matter releases carbon dioxide gas into the atmosphere. When the carbon dioxide gas dissolves in water, the following processes take place.



At room temperature and pressure, the mole fraction of carbon dioxide gas in normal dry air is about 3.5×10^{-4} .

(a) Using relevant data given above,

- (i) calculate the solubility of carbon dioxide gas, in mol dm^{-3} , in water at a pressure of 101.3 kPa.

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[2]

- (ii) calculate the equilibrium concentration of carbonic acid, H_2CO_3 .

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[1]

- (b) Hence, calculate the pH of water when carbon dioxide gas in the atmosphere is dissolved in it (ignore the effects of K_{a2} on the pH).

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[2]

- (c) Most waters contain some decaying organic matter. Explain, without calculations, how this will affect the pH of the water.

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.....[3]

- (d) Observations from an experimental study showed that in water, each molecule of carbonic acid is hydrated with an average of 3 water molecules. No two water molecules forms bond with atoms on carbonic acid that are bonded to each other.

- (i) Suggest the type of interactions formed between carbonic acid and water molecules.

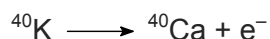
.....[1]

- (ii) Draw the structure of a molecule of carbonic acid. Show clearly how the 3 water molecules are arranged around the carbonic acid molecule.

[3]

[Total: 12]

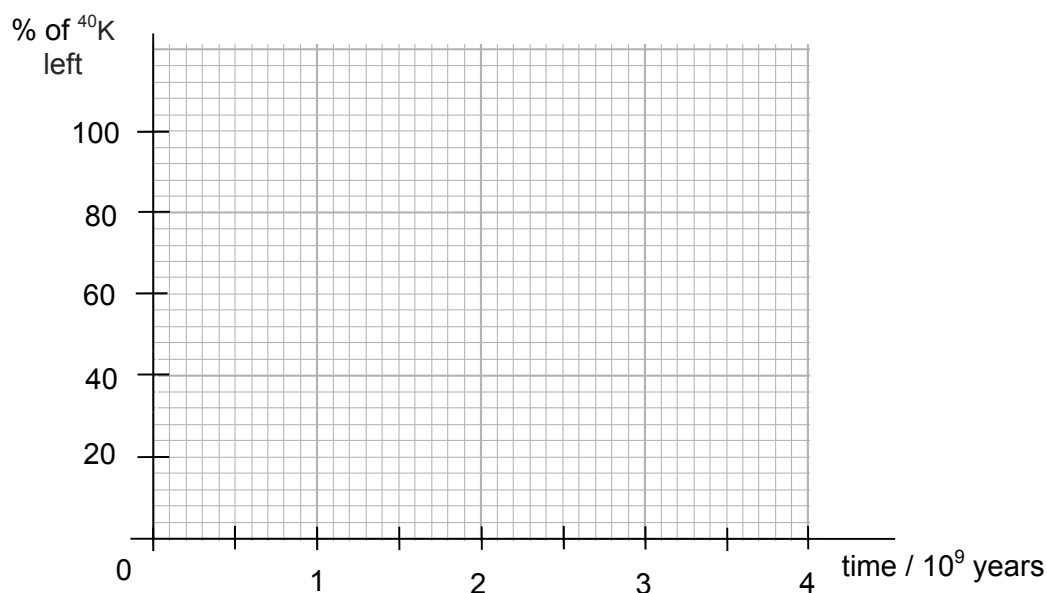
- 4 Potassium-40, ^{40}K , is a natural radioactive element found in the Earth's mantle which produces high amount of radiogenic heat as it decays. ^{40}K undergoes radioactive decay to form ^{40}Ca according to the equation below.



The following measurements relate to the decay of ^{40}K .

percentage of ^{40}K left	time / years
100	0
80	0.45×10^9
52	1.25×10^9
32	2.10×10^9
20	3.00×10^9
13	4.00×10^9

- (a) (i) Plot these measurements on the grid below.



[1]

- (ii) Show clear working to justify that the order of reaction of this decay is first order.

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.....[2]

- (b) In a sample of mineral mica whose age was to be determined, the calcium produced as a result of the above decay was estimated to be 128.5×10^{18} atoms, when the ^{40}K content was 77.1×10^{18} atoms.

(i) What is the number of ^{40}K atoms present originally in the mica sample?

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.....[1]

(ii) Hence, estimate the age of the mica sample.

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.....
.....
.....[1]

- (c) Elements can be artificially produced via nuclear reactions.

During the process, protons or neutrons are injected 'like bullets' at the nucleus of an atom in making the synthetic element.

Suggest why neutrons are more effective than protons as nuclear bullets.

.....
.....[1]

- (d) **L** and **M** are atoms of elements in the Periodic Table. Some data concerning the particles derived from **L** and **M** are given in the table below.

Particle	Mass number	Number of		
		protons	electrons	neutrons
L ²⁺	111		47	
M ²⁻	51		27	

- (i) Complete the above table. [2]

- (ii) Use the *Data Booklet* to predict a value for the energy required to form one mole of gaseous **L**²⁺ ions from one mole of its gaseous atoms.

Explain your answer.

.....

[2]

- (iii) When a beam of ⁴⁰Ca²⁺ particles travels through a uniform electric field which is at right angles to its direction of travel, it is deflected at an angle of +10.0°.

Determine the angle of deflection of a beam of **M**²⁻ particles if it travels at the same speed through the same electric field.

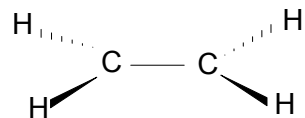
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[2]

[Total: 12]

5 (a) The bonding in ethene and benzene may be described as a mixture of σ and π bonding.

(i) Each carbon atom in ethene forms three σ bonds as shown below.



On the diagram, sketch the π bond that is also present in ethene. [1]

(ii) Explain why the π bond in ethene is described as a *localised* π bond.

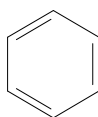
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[1]

(iii) The molecules of benzene is a regular hexagon in which the π electrons are described as being delocalised.

By using benzene as an example, explain, with the aid of a diagram, what is meant by the term *delocalised* π bond.

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[2]

(iv) Kekulé proposed the following structure for benzene.



The enthalpy change of hydrogenation of cyclohexene, as shown, is -121 kJ mol^{-1} .

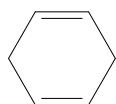


Based on this value for cyclohexene and Kekulé's structure, it is possible to calculate that the enthalpy change of hydrogenation of benzene to be -363 kJ mol^{-1} .

Explain the difference between this calculated value and the actual value for the enthalpy change of hydrogenation of benzene which is -209 kJ mol^{-1} .

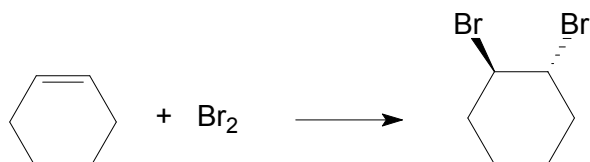
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.....
.....[2]

- (b) Cyclohexa-1,4-diene displays reactivity typical of alkenes. Its structure is shown.



cyclohexa-1,4-diene

When a bromine molecule adds across a C=C double bond, the two bromine atoms bond to **opposite** faces of the molecule as illustrated using cyclohexene.



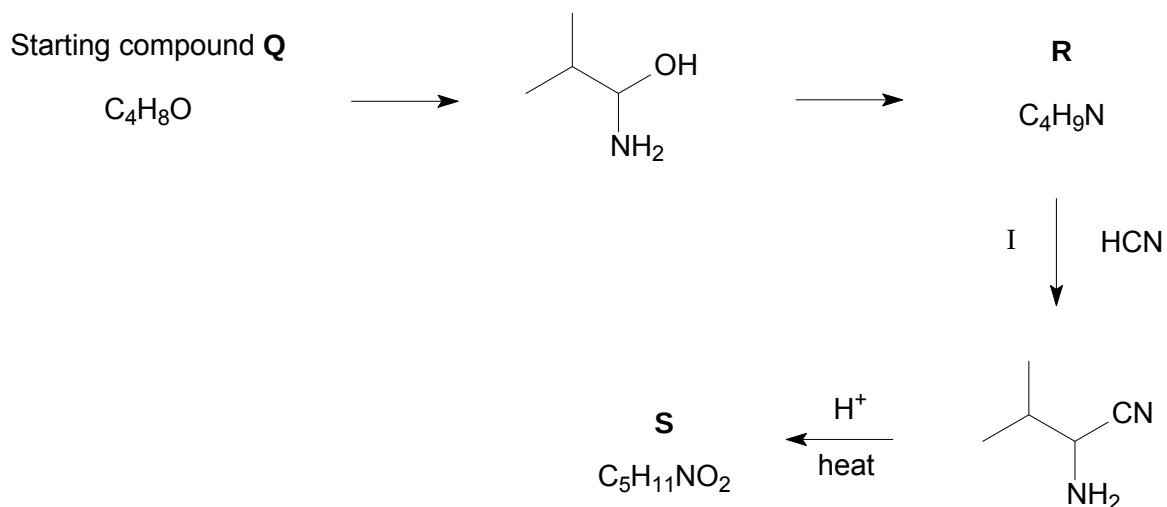
When one molecule of cyclohexa-1,4-diene reacts with **two** molecules of bromine, a mixture of **three** stereoisomers are formed. Draw the structures of the stereoisomers formed.

You may find it useful to show the six-membered carbon ring as a hexagon in the plane of the paper and use wedged and dashed bonds to bromine to show the stereochemistry.

[3]

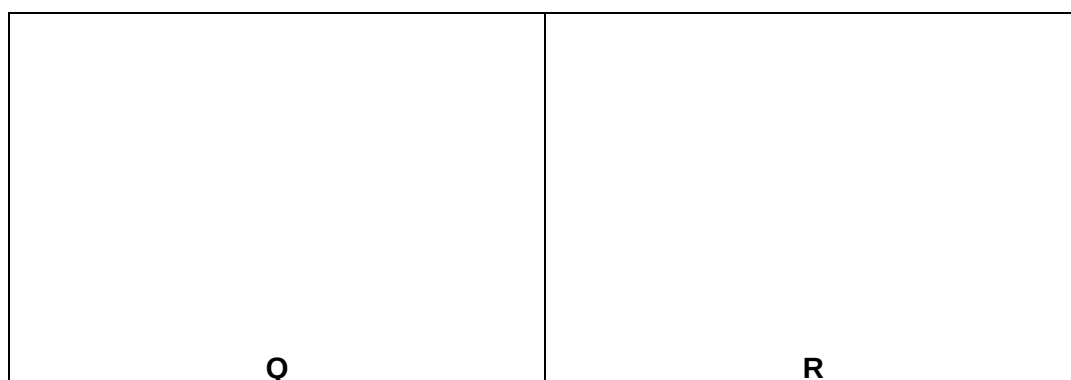
[Total: 9]

- 6 The Strecker amino acid synthesis, developed in 1850 by Adolph Strecker, is a series of chemical reactions to synthesise an α -amino acid. Compound **S** is one such α -amino acid.



Compound **Q** does not react with sodium metal and compound **R** does not decolorise bromine water.

- (a) (i) Suggest the structures of compounds **Q** and **R**.



[2]

- (ii) What type of reaction is occurring in step I?

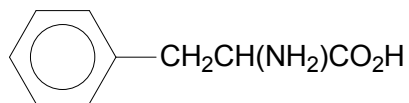
.....[1]

S is a neutral solid and has a high melting point.

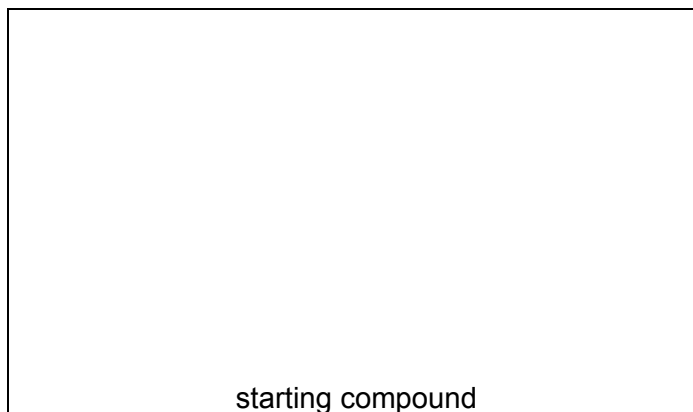
- (iii) Suggest the type of structure present in **S** and draw a formula of **S** which represents this structure.

.....[1]

- (iv) Suggest the structure of the starting compound if the following α -amino acid is to be synthesised via the Strecker synthesis.



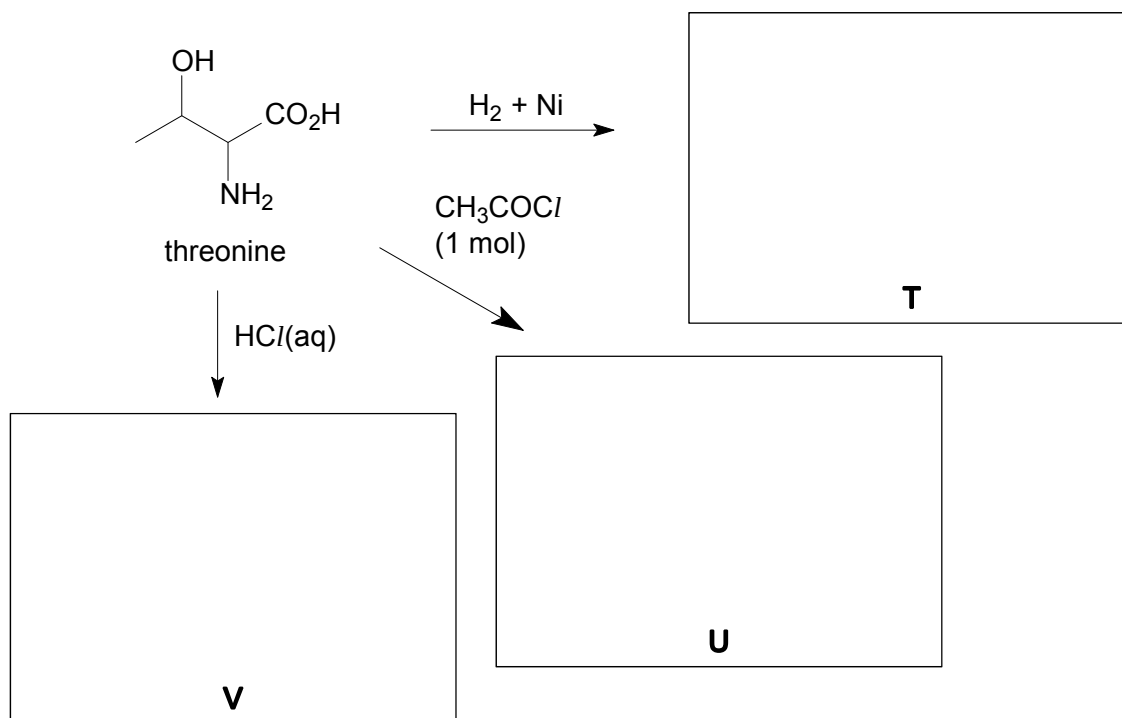
phenylalanine



[1]

- (b) Another α -amino acid, threonine, was treated with the respective reagents in the following three reactions.

Draw the structures of the products formed in the boxes provided. If nothing is formed, write 'no reaction'.



[3]

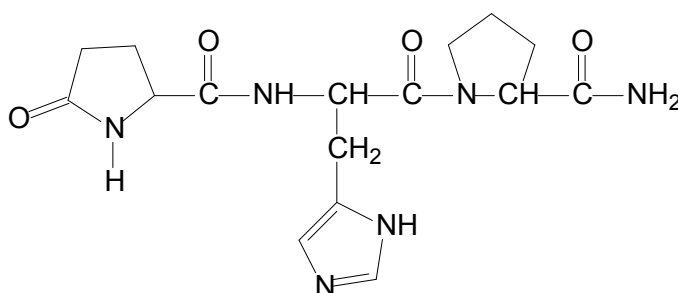
- (c) Some diseases are caused by a mutation in the DNA base sequence which results in one amino acid being replaced by another during protein synthesis.

Suggest, in terms of the type of interactions, how the protein structure will change when a phenylalanine residue is replaced by a threonine residue.

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[2]

- (d) Pyroglutamyl histidyl prolinamide is a thyrotropin-releasing hormone (TRH) that stimulates the secretion of thyrotropin, which in turn stimulates the secretion of thyroid hormones, the regulators of the body's metabolism.



pyroglutamyl histidyl prolinamide

- (i) How many **different** amino acid residues are present within each TRH molecule?

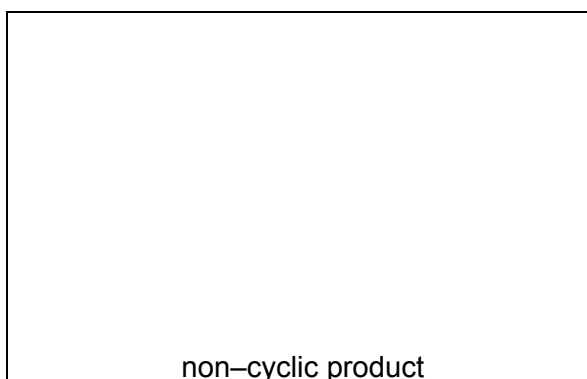
.....[1]

The activity of pyroglutamyl histidyl prolinamide is destroyed by heating it under reflux with 6 mol dm^{-3} hydrochloric acid for a prolonged period.

- (ii) State the type of reaction occurring.

.....[1]

- (iii) Draw the structure of the **non-cyclic** product formed under these conditions.



[1]

[Total: 13]